

Debdip Bandyopadhyay

Senior AI & Automation Engineer | Intelligent Verification & Systems Quality

Kolkata, India | +91-8285578544 | debdip1992@outlook.com | linkedin.com/in/debdiparvr | github.com/debdipARVR

PROFESSIONAL SUMMARY

Senior AI & Automation Systems Engineer with **11+ years of experience** architecting intelligent verification workflows, agentic AI pipelines, and high-throughput regression automation across **Generative AI platforms, High-Performance GPU Computing, and Safety-Critical Aerospace Systems**. Core member of Cognizant's GenAI CoE, pioneering LLM-as-a-Judge evaluators backed by **NVIDIA NIM**, automated regression anomaly detection, and Dockerized parallel execution grids slashing regression cycles by **~40%**. Rigorous engineering discipline developed across mission-critical avionics verification at **The Boeing Company**. Master's degree from **IIT Jodhpur (School of AI & Data Science)** with hands-on compute on **NVIDIA DGX A100 GPU clusters** and independent published benchmark research evaluating frontier foundation models including **NVIDIA Nemotron-3 120B** (Elsevier SSRN DOIs: [10.2139/ssrn.7374319](https://ssrn.com/abstract=10.2139/ssrn.7374319), [10.2139/ssrn.7374338](https://ssrn.com/abstract=10.2139/ssrn.7374338), [10.2139/ssrn.7374378](https://ssrn.com/abstract=10.2139/ssrn.7374378) & CERN Zenodo DOIs: [10.5281/zenodo.22177214](https://zenodo.org/record/22177214), [10.5281/zenodo.22113940](https://zenodo.org/record/22113940)).

TECHNICAL SKILLS

- **AI Quality, LLM Evaluation & Agents:** NVIDIA NIM API, NVIDIA Nemotron, Agentic AI Frameworks, DeepEval, LLM-as-a-Judge, RAG Validation, PyTorch, Scikit-learn, Non-Deterministic Output Verification, Semantic Faithfulness Scoring.
- **Verification Automation & Systems:** Python (Advanced), C++, Linux / Bash, Pytest, Pytest-BDD, Playwright, API Testing (Requests, Postman), Parallel Sharding (pytest-xdist), High-Throughput Failure Triage, Anomaly Detection.
- **CI/CD, Infrastructure & Compute:** Docker, Docker Compose (Multi-Container Test Grids), Azure DevOps Pipelines, Git, Allure Merged Reporting, NVIDIA DGX A100 Clusters, Zero-Escape Quality Gates.
- **Methodologies & Systems Rigor:** Safety-Critical Systems Verification (Aerospace / DO-178C standards), Regression Profiling & Bottleneck Optimization, Continuous Model Benchmarking, Statistical Evaluation.

OPEN-SOURCE AI SYSTEMS & GITHUB PROJECTS

NVIDIA NIM-Powered LLM Verification Framework

github.com/debdipARVR/chatbot_tests_deepeval_bitnet

[Custom DeepEval LLM-as-a-Judge evaluator backed by NVIDIA NIM API ([integrate.api.nvidia.com](https://api.nvidia.com))]

- Engineered custom **NvidiaLLM** evaluator integrating NVIDIA NIM microservices with DeepEval's G-Eval framework.
- Automated semantic verification, factual correctness matching, and hallucination scoring on non-deterministic LLM responses.

Distributed Dockerized Parallel Test Grid

github.com/debdipARVR/debdipARVR-Testing

[Multi-container parallel test architecture with dynamic execution sharding and merged Allure reporting]

- Architected containerized parallel test runners orchestrating multi-container execution nodes via Docker Compose.
- Implemented automated test artifact consolidation and multi-container report aggregation for high-scale regression suites.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Apr 2021 — Present

Senior AI Quality Engineer / Senior SDET — GenAI CoE Team

Kolkata, India

- **GenAI CoE Quality Ownership:** Core member of the GenAI Center of Excellence (CoE) team, driving end-to-end verification strategies and collaborating with NLP researchers, data scientists, and cloud architects to deliver production-grade AI solutions.
- **NVIDIA NIM & DeepEval Pipelines:** Built automated AI evaluation suites for LLM and RAG applications using DeepEval and NVIDIA NIM, systematically validating non-deterministic outputs across correctness, faithfulness, and semantic congruence.
- **Intelligent Triage & Anomaly Isolation:** Developed automated failure triage and analysis pipelines to parse regression results, detect performance anomalies, and accelerate root-cause isolation across multi-stage release cycles.
- **High-Throughput Parallel Automation:** Architected enterprise-grade Python test automation frameworks leveraging parallel execution (pytest-xdist) and container sharding, slashing regression execution cycle times by **~40%**.
- **CI/CD Quality Gates:** Integrated automated verification execution into Azure DevOps CI/CD pipelines, enforcing strict quality gates and regression tracking on every deployment.
- **Cross-Domain Leadership:** Mentored 10+ automation engineers on test architecture, AI evaluation protocols, and code review standards.

The Boeing Company

Oct 2018 — Apr 2021

Software Engineer — Test Automation

Bengaluru, India

- **Safety-Critical Systems Verification:** Engineered automated verification pipelines for mission-critical aerospace software operating under strict zero-tolerance engineering standards.
- **High-Volume Regression Optimization:** Maintained, profiled, and executed high-consequence regression test suites of 2,000+ automated test cases across multi-phase release cycles, eliminating release escapes and optimizing execution bottlenecks.
- **Mentorship & Technical Rigor:** Mentored junior test engineers on coding standards, automated test hygiene, and architectural code reviews.
- **Recognition:** Awarded **Boeing Pride Award** and **Star Award** (2019) for exceptional verification delivery under compressed schedules.

PROFESSIONAL EXPERIENCE (CONTINUED)

Accenture LLP

Oct 2014 — Oct 2018

Test Automation Engineer

Kolkata / Noida, India

- **Automated Regression Pipelines:** Developed modular, data-driven automation frameworks across high-volume distributed application suites, automating system integration and regression verification workflows.
- **Tooling Engineering & Cost Optimization:** Designed and deployed an in-house One Click Capture test automation utility, eliminating manual capture overhead and reducing third-party tooling expenses by ~USD 8,000 annually.
- **Requirement Decomposition:** Collaborated with system architects and domain leads to translate complex system specifications into automated verification test suites.

PUBLICATIONS & AI RESEARCH PREPRINTS

Multi-Scale 4-Pass Sentence Cloze Infilling with Pass-Adaptive Dynamic Gating for Robust AI Text Detection

[Zero-shot black-box framework achieving 99.58% AUROC on RAID (ACL 2024), 99.98% on HC3, and 0.83% FPR on formal human academic prose across a 240-essay dual corpus via Kuhn-Munkres S_3 bipartite matching and continuous sigmoidal gating]

Elsevier SSRN: ssrn.com/abstract=7374319 | SSRN DOI: 10.2139/ssrn.7374319 | CERN Zenodo DOI: 10.5281/zenodo.22177214 |

Benchmarks: github.com/debdipARVR/cloze_congruence_reproducibility_benchmark_kit

ClozeCongruence: Model Fingerprinting and Traceback Attribution in LLMs via Reverse Cloze Resonance

[Foundation model provenance traceback achieving 100.00% Top-1 attribution accuracy (Macro- $F_1=1.0000$) across 6 frontier LLMs including NVIDIA Nemotron-3 120B, GPT-4o, Claude 3.5, and Gemini 3.7 with +82.21% self-model resonance gap and 12D micro-stylometrics]

Elsevier SSRN: ssrn.com/abstract=7374338 | SSRN DOI: 10.2139/ssrn.7374338 | CERN Zenodo DOI: 10.5281/zenodo.22113940 | Interactive

Demo: huggingface.co/spaces/DebdipCS/cloze-congruence-ai-forensics-hub-v1

The Cloze Scaling Laws: Prober Parameter Dynamics, Native Blank-Infilling Architecture, and Traceback

[Systematic empirical scaling study establishing monotonic prober recovery (12B to 120B) and GLM-5.2 native blank-infilling inductive advantage (+51.16% separation gap, Cohen's $d=5.62$, $p=2.87e-15$) across international SOTA benchmarks]

Elsevier SSRN: ssrn.com/abstract=7374378 | SSRN DOI: 10.2139/ssrn.7374378 | CERN Zenodo DOI: 10.5281/zenodo.22158286 | Author

Portfolio: ssrn.com/author=12882248

EDUCATION

Master of Technology (M.Tech) in AR & VR

Jul 2022 — Dec 2025

Indian Institute of Technology Jodhpur (School of Artificial Intelligence & Data Science)

Jodhpur, India

- **Key Coursework:** Deep Learning, Generative AI, Natural Language Processing (NLP), MLOps & DLOps, C++, Data Structures & Algorithms, Reinforcement Learning Simulation.
- **High-Performance Compute:** Hands-on model training and high-throughput simulation on NVIDIA DGX A100 GPU clusters.

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Apr 2010 — Jun 2014

Greater Noida Institute of Technology (AKTU University)

Greater Noida, India

Higher Secondary (Class XII) — CBSE [PCMB]

2010

Kendriya Vidyalaya No. 1, Salt Lake (KV No. 1 SLK)

Kolkata, India

CERTIFICATIONS

- Postman API Testing Path (v12)
- ASDA-MIT Certification
- Google Crash Course on Python (Coursera)